SOMAIYA
VIDYAVIHAR UNIVERSITY

23.05.2025(E)

Semester: January 2025 – April 2025		
Maximum Marks: 50	**Examination: End-Semester Examination**	**Duration: 2 Hrs.**
Programme code: 03 **Programme: EXTC**	**Class: SY BTech**	**Semester: III (SVU 2023)**
Institute/School/Department: K. J. Somaiya School of Engineering	**Name of the department:** EXTC	
Course Code: 216U03C303	**Name of the Course: Digital Logic Design**	
Instructions: 1) Draw neat diagrams 2) All questions are compulsory 3) Assume suitable data wherever necessary		

Que. No.	Question Statement	Max. Marks
Q.1	Attempt any **two**	
i)	Answer the following 1. Perform the following using 2's complement method $(34-21)_{10}$ using 8 bit representation 2. Add $(37)_{10}$ and $(41)_{10}$ in BCD	05
ii)	Design 4:1 multiplexer using basic gates.	05
iii)	Design and Implement 1 bit binary Full Adder using basic gates.	05
Q.2	Attempt any **one**	10
i)	For the following given K-map : 1) Identify the type of the function. 2)Draw Truth Table 3) What will be the minimized equation of the given function? 4)Realize the minimized equation using basic gates 5) Realize the minimized equation using universal gates.	10

A \ B,C	00	01	11	10
0	0	0	1	0
1	0	1	1	1

Que. No.	Question Statement	Max. Marks
ii)	Implement the following function $F(A,B,C,D) = \sum m(0,1,5,8,10,13,15)$ using: 1) 16:1 Multiplexer	05
	2) only one 8:1 multiplexer	05

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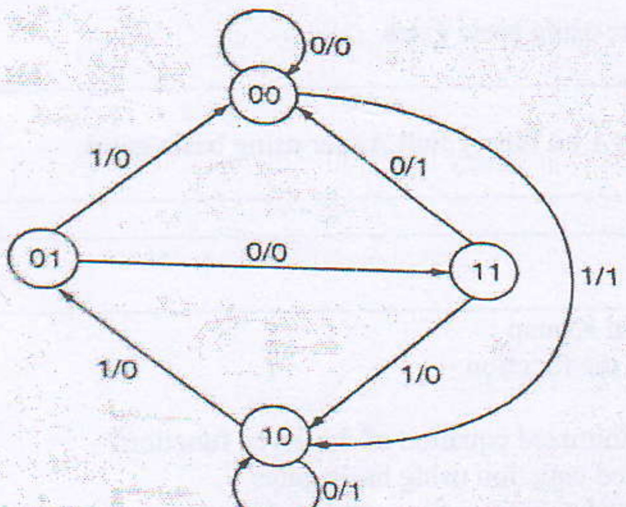
Q.3	Attempt any **one**	**10**
i)	Design and Implement 3 bit Asynchronous Up counter using JK FF. (Assume Negative edge triggered clock, Active low RESET and CLEAR pins)	**10**
ii)a)	Implement and draw a 3-bit ring counter using D FF.	**05**
ii)b)	Draw diagrams for JK FF converted into 1) T FF 2) D FF. (Assume Clock is negative edge triggered type)	**05**
Q.4	Attempt the following	**10**
	Design and implement a synchronous sequential circuit using D FFs for the following state diagram.  	
Q.5	Attempt the following	**10**
	What is the need of DAC in digital circuits? Explain 4 bit R-2R DAC.	